

MODELLING DIELECTRIC PERMITTIVITY OF WATER CONFINED BETWEEN CHARGED LIPID LAYERS WITH A MODIFIED POISSON-BOLTZMANN MODEL

Ludovic GARDRE¹, Swen HELSTROFFER², Pierre MULLER², Fabrice THALMANN², Thierry CHARITAT², Laurent JOLY¹, Claire LOISON¹

¹ Université Claude Bernard Lyon 1, CNRS, ILM, UMR 5306, Villeurbanne, France

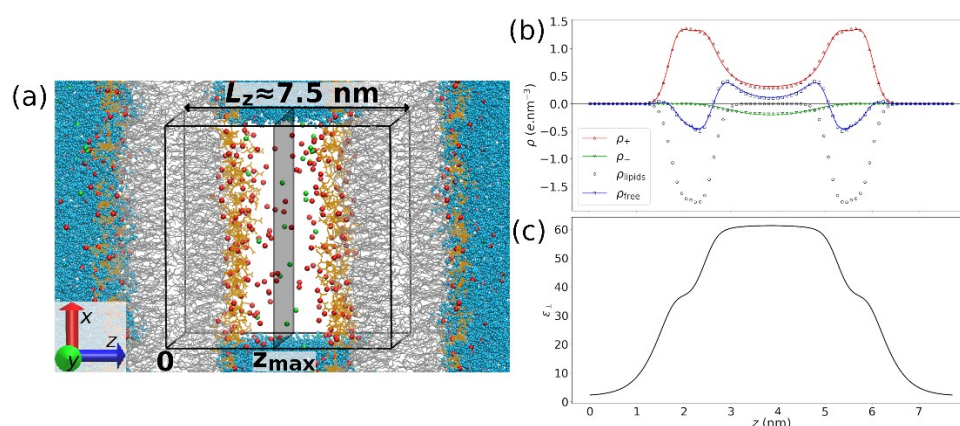
² Université de Strasbourg, CNRS, ICS, UPR 22, Strasbourg, France

(ludovic.gardre@univ-lyon1.fr)

Keywords: Poisson-Boltzmann, Dielectric permittivity, Strong coupling, Electrostatics

ABSTRACT

Lipid membranes play key roles in biological systems. Electrostatic interactions often govern nearby dynamics and can even affect membrane structure: like-charged lipid layers immersed in multivalent ion solutions can attract each other [1]. This counter-intuitive effect is explained by strong coupling driven by ion correlations. Recent experiments [2] show that similar attraction occurs with monovalent ions, possibly due to reduced water permittivity under confinement. To assess this explanation, we investigate the water permittivity using molecular dynamics simulation and modelling.



We develop a modified Poisson-Boltzmann model with spatially varying permittivity and explicit Born solvation energy for ions. We use it to extract effective permittivity profiles (Fig. c) from MD simulations of charged lipid layers (Fig. a). The model uses this permittivity to compute ionic charge densities (lines in Fig. b), which are in good agreement with those from the simulations (markers in Fig. b). Our approach could be applied to various systems where electrolytes are confined between charged interfaces, such as polyelectrolytes, biological molecules or clays.

REFERENCES

- 1 K. Komorowski, et al., *Biophysical Journal*, **114**, 1908-1920 (2018), DOI : [10.1016/j.bpj.2018.02.040](https://doi.org/10.1016/j.bpj.2018.02.040).

2 T. Mukhina, et al., *J Phys Chem Lett.*, **10(22)**, 7195-7199 (2019), DOI : [10.1021/acs.jpclett.9b02804](https://doi.org/10.1021/acs.jpclett.9b02804).